Q-Pathformer - Path 1,2, & 3 Run Results

Validated Source, Lifecycle, and Controlled Local Run Results Across the Three Dataset Paths

Compatible with Access-PoD Governance and Q-Pathformer Execution Environments

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Document Scope

This working results document records the current source material, execution approach, lifecycle outcome, and controlled local validation results for the Q-Pathformer three-path release structure.

It summarizes the currently validated Path 1 and Path 2 runs, including source-first and signal-first testing approaches where applicable, and preserves a placeholder structure for Path 3 completion.

This document is descriptive rather than normative. It does not replace the governance or lifecycle definitions established in APOD-TR-007 and APOD-TR-008, and any local fine-tuning activity described here remains controlled validation unless separately authorized.

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1. Introduction

This document presents the validated execution results for the **Q-Pathformer Enterprise v1.0 release**, covering the three core dataset pathways used in the system:

- Path 1 — TR corpus reference path
- Path 2 — Access-Mode lifecycle path
- Path 3 — external and custom dataset adapter path

The purpose of this document is to provide a **clear, structured summary of how each path operates in practice**, what lifecycle stage it reaches, and how the results should be interpreted within the governed system.

Unlike the technical runtime logs and detailed execution records, this document is written as a **release-facing interpretation layer**. It simplifies the underlying runs into a readable format while preserving the key outcomes and constraints.

This document should be read alongside:

- APOD-TR-008 — execution routing model
- APOD-TR-009-CSC — CSC interpretation model
- QPATH-RUN-001 (.md) — detailed execution trace and technical results

The .md results file provides the full technical detail of each run, including dataset construction, lifecycle transitions, and training validation outputs.
This document focuses on **summarizing those results in a structured and interpretable form**.

1.1 Scope

This document describes:

- the validated results from each dataset path in the v1.0 release
- the lifecycle state reached by each path
- the role each path plays within the system
- and the boundaries that apply to training and dataset usage

This document is **descriptive and non-normative**.

It does not:

- define governance rules
- modify lifecycle semantics
- or grant training authorization

All results described here remain subject to:

- lifecycle containment
- governance controls
- and human review

1.2 Key Interpretation Principle

Across all paths, a critical constraint applies:

No output in this release should be interpreted as unrestricted training authorization.

The system separates:

- dataset preparation
- lifecycle progression
- and training authorization

This ensures that:

- execution results do not imply approval
- dataset structure does not imply readiness
- and lifecycle state does not imply correctness

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2. Overview of the Three Paths

The Q-Pathformer Enterprise system operates through three distinct dataset pathways. Each path serves a different role in the overall system.

Path 1 — TR Corpus Reference Path

- internal reference pathway
- transforms technical report material into governed datasets
- produces a complete training-ready package
- used for controlled local validation

Path 2 — Access-Mode Lifecycle Path

- governance-driven lifecycle pathway
- starts from Access-Mode source artifacts
- progresses through validation, promotion, and freeze

- produces a controlled STATE_1 dataset
- primary example of lifecycle governance

Path 3 — External / Custom Adapter Path

- onboarding pathway for external or client data
- normalizes input into governed dataset structure
- remains contained in STATE_3
- does not promote to training by default

Path Relationship

These paths represent three different system functions:

- Path 1 → **reference training system**
- Path 2 → **governed lifecycle system**
- Path 3 → **external onboarding system**

Together, they demonstrate how:

- source material
- runtime signals
- and external datasets

can be handled within a **single governed lifecycle framework**.

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3. Path 1 — TR Corpus Reference Path

3.1 Purpose

Path 1 is the internal reference pathway for the Q-Pathformer system.

It transforms the project's technical report material into:

- structured signals
- governed datasets
- and a controlled training-preparation package

This path demonstrates how internal source material can be carried through the full dataset lifecycle into a usable and validated dataset structure.

3.2 Source and Entry Point

Path 1 uses the internal reference corpus included in the release:

- path_1_tr_corpus_reference\01_source\tr_docs
- path_1_tr_corpus_reference\01_source\readme_refs

The corpus contains:

- technical report documents
- Access-Mode reference material
- structured governance definitions, rules, and constraints

The signal extraction layer produced:

- **16 documents processed**
- **846 structured signals extracted**

3.3 Execution Flow

The Path 1 pipeline is deterministic and does not depend on runtime model signal generation:

1. build corpus snapshot
2. extract structured signals
3. construct governed dataset
4. perform dataset filtering and normalization
5. pass STATE_2 review
6. execute promotion
7. perform STATE_1 handoff
8. prepare training package

This ensures:

- traceability from source to dataset
- strict governance enforcement
- consistent dataset structure

3.4 Result Summary

The validated Path 1 run produced:

- dataset_version: checkpoint_v4_strict
- source_signal_rows: 1241
- train_rows: 1117
- validation_rows: 124

The dataset successfully passed:

- STATE_2 review (approved)
- STATE_1 promotion (approved)
- STATE_1 handoff (approved)

3.5 Lifecycle Outcome

Path 1 reaches:

- **STATE_1 — training-preparation stage**

This means:

- the dataset is fully constructed and governed
- lifecycle checks have been passed
- the dataset is ready for controlled validation

3.6 Controlled Training Validation

Following STATE_1 handoff, a controlled local LoRA validation run was executed.

Run details:

- model base: TinyLlama 1.1B
- training mode: local LoRA
- record_count: 1117

Important:

- this run occurs **after lifecycle completion**
- it does **not change lifecycle state**
- it is used for **validation only**

3.7 Critical Constraint

Even with successful validation:

- training is **not globally authorized**
- dataset usage remains **controlled**
- lifecycle state does **not imply permission**

The dataset posture is explicitly:

- **non-authoritative**
- **training-preparation only**

3.8 Interpretation

Path 1 is the strongest **internal reference dataset pathway** in the v1.0 release.

It demonstrates:

- full transformation from source documents to dataset
- strict lifecycle governance enforcement
- successful progression to STATE_1
- controlled execution validation

It is important to distinguish:

- lifecycle progression (STATE_1)
- from training execution (validation only)

This separation ensures that:

- dataset readiness does not imply usage
- execution does not imply authority
- and governance remains external to the system

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4. Path 2 — Access-Mode Lifecycle Path

4.1 Purpose

Path 2 is the primary **governed lifecycle pathway** in the Q-Pathformer system.

It demonstrates how Access-Mode artifacts are:

- captured

- preserved through freeze discipline
- transformed into governed signals
- and progressed through lifecycle stages

This path provides the strongest example of how the system enforces:

- provenance
- lifecycle control
- and governance-aware dataset progression

4.2 Path 2 Structure

Path 2 operates in two distinct modes:

- **Source-first lifecycle path** (primary release example)
- **Runtime signal path (gpt-oss:20b option)**

Both use the same lifecycle model but differ in how source data enters the system.

4.3 Path 2 — Source-First Lifecycle Path

4.3.1 Entry Point

The source-first path begins from a frozen Access-Mode bundle:

- path_2_access_mode_canonical\01_source\2026-04-11--FREEZE-20260412-000400Z

This bundle contains:

- canonical session records
- canonical module records
- freeze note and hash evidence
- full provenance from the Access-Mode workflow

This ensures the run starts from a **true ID_3 frozen source**, not generated signals.

4.3.2 Execution Flow

The source-first run follows a complete lifecycle sequence:

1. ingest frozen source bundle
2. bridge source into governed signals

3. export grouped candidate dataset
4. perform STATE_2 review
5. execute controlled promotion
6. perform dataset handoff and freeze
7. update registry
8. build training package
9. run controlled local validation

This sequence preserves:

- source integrity
- lifecycle traceability
- and governance constraints

4.3.3 Result Summary

The validated run produced:

- candidate_count: 4
- source_record_count: 16
- routing_decision: SEND_TO_STATE_2_REVIEW

Lifecycle progression:

- STATE_2 review passed
- promotion passed
- handoff passed
- freeze passed
- registry update passed

Training validation:

- model base: TinyLlama 1.1B
- records_loaded: 4
- train_loss: 3.7577
- train_loss_final: 3.2855

4.3.4 Lifecycle Outcome

Path 2 (source-first) reaches:

- **STATE_1 — controlled training-preparation stage**

This confirms that:

- the dataset passed all lifecycle gates
- governance controls were enforced
- and a valid training package was produced

4.3.5 Critical Constraint

Even though Path 2 reaches STATE_1:

- fine-tuning is **not globally authorized**
- training remains **controlled validation only**

This path demonstrates:

- lifecycle execution
- not unrestricted training permission

4.3.6 Interpretation

The source-first Path 2 run is the **strongest governed lifecycle example** in the release.

It demonstrates:

- full provenance from source to dataset
- enforcement of review and promotion gates
- deterministic lifecycle progression
- preservation of freeze and registry discipline

This path is the **primary evidence source for lifecycle behavior** in the system.

4.4 Path 2 — Runtime Signal Option (gpt-oss:20b)

4.4.1 Purpose

The runtime option provides a **live signal-entry pathway** using model-generated output instead of a pre-existing source bundle.

It allows:

- rapid signal generation
- partial lifecycle validation
- testing of runtime-to-review flow

4.4.2 Entry Point

This path starts from runtime signal generation:

- path_2_access_mode_canonical\02_signals\runtime_generated
- model: gpt-oss:20b

Unlike the source-first path:

- it does **not use the frozen source bundle**
- it creates signals dynamically at runtime

4.4.3 Execution Behavior

The runtime path performs:

- signal generation from model output
- placement into STATE_3
- candidate export grouped by session
- optional review and validation

The lifecycle is identical, but:

- the source layer is generated instead of pre-frozen

4.4.4 Result Summary

Validated runtime result:

- candidate_count: 1
- routing_decision: SEND_TO_STATE_2_REVIEW
- STATE_2 review passed

However:

- promotion was **held** due to insufficient candidate count

Constraint:

- minimum candidates required for promotion = 4

4.4.5 Lifecycle Outcome

Runtime path reaches:

- STATE_3 → STATE_2 (review approved)
- **does not reach STATE_1**

4.4.6 Interpretation

The runtime option is:

- a **valid operational pathway**
- suitable for signal testing and lifecycle validation

However:

- it is not equivalent to the source-first path
- it requires sufficient grouped signals to progress
- it does not demonstrate full lifecycle completion by default

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4.5 Overall Interpretation of Path 2

Path 2 demonstrates the system's **governed lifecycle capability**.

- The **source-first path** shows full lifecycle completion
- The **runtime path** shows live signal handling with constrained promotion

Together, they confirm:

- lifecycle routing is functioning correctly
- promotion is controlled and gated
- and dataset progression is not automatic

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5. Path 3 — External and Custom Dataset Adapter Path

5.1 Purpose

Path 3 provides the **external and custom dataset onboarding pathway** within the Q-Pathformer system.

It is used to:

- intake external governed bundles
- ingest client-origin structured datasets
- normalize input into governed dataset structures
- and place all outputs into controlled lifecycle containment

Unlike Path 1 and Path 2, Path 3 does **not progress datasets into training preparation by default**.

Its role is to ensure that external data can be:

- safely introduced
- structurally validated
- and governed from the point of entry

5.2 Path 3 Structure

Path 3 currently supports two validated intake branches:

- **External governed bundle intake**
- **Client structured dataset intake**

Both branches:

- normalize input into a common dataset format
- preserve provenance and structure
- and remain contained within the lifecycle

5.3 Path 3 — External Governed Bundle Intake

5.3.1 Entry Point

This branch begins with an inbound governed bundle:

- path_3_external_custom_adapter\01_source\inbound\QPDS-PACKAGE-001

This bundle contains:

- canonical module records
- canonical session records
- freeze note
- hash manifest

Although structurally similar to an ID_3 freeze bundle, it is treated here as an **external input source**, not an internal lifecycle artifact.

5.3.2 Execution Flow

The ingestion process performs:

1. validation of bundle structure and hash integrity
2. normalization into dataset-compatible records
3. preservation of provenance and source structure
4. creation of a governed onboarding package

This ensures that external data is:

- interpreted consistently
- structurally aligned with system expectations
- and prepared for lifecycle handling

5.3.3 Result Summary

The validated run produced:

- source_file_count: 6
- normalized_record_count: 6
- module_record_count: 2
- session_record_count: 2
- hash_verification_status: ok

All records were successfully:

- validated
- normalized
- and written to the onboarding dataset

5.3.4 Lifecycle Outcome

This branch remains in:

- **STATE_3 — contained onboarding stage**

It does not:

- enter validation review
- progress to promotion
- or reach training preparation

5.3.5 Interpretation

The external bundle branch demonstrates:

- controlled intake of governed external artifacts
- preservation of source integrity and hash validation
- successful normalization into system-compatible structures

It is a **complete onboarding example**, not a lifecycle progression example.

5.4 Path 3 — Client Structured Dataset Intake

5.4.1 Entry Point

This branch begins from a structured client dataset:

- path_3_external_custom_adapter\01_source\inbound\CLIENT-DATASET-EXAMPLE-001\client_governance.json

This represents:

- a simplified external dataset format
- typical of client-origin structured data

5.4.2 Execution Flow

The ingestion process performs:

1. validation of input JSON structure
2. normalization into dataset rows
3. assignment of governance fields
4. creation of onboarding dataset package

This process aligns client data with:

- the same schema
- lifecycle model
- and governance structure used internally

5.4.3 Result Summary

The validated run produced:

- source_file_count: 1
- normalized_record_count: 4
- client_governance_record_count: 4
- hash_verification_status: ok

All records were successfully:

- validated
- normalized
- and prepared for lifecycle handling

5.4.4 Lifecycle Outcome

This branch also remains in:

- **STATE_3 — contained onboarding stage**

It does not:

- enter validation review
- progress to promotion
- or reach training preparation

5.4.5 Interpretation

The client dataset branch demonstrates:

- intake of simplified external data formats
- structural normalization into governed datasets
- alignment with lifecycle containment rules

It confirms that:

- client-origin data can be onboarded safely
- without bypassing governance controls

5.5 Shared Lifecycle and Constraints

Both Path 3 branches operate under the same constraints:

- lifecycle_state = STATE_3
- routing_decision = HOLD_IN_STATE_3
- fine_tuning_allowed = false
- external_training_allowed = false
- human_review_required = true

This ensures that:

- all external data remains contained
- no dataset is promoted without review
- and no training occurs by default

5.6 Interpretation

Path 3 represents the system's **controlled onboarding layer**.

It demonstrates:

- safe intake of external and client datasets
- normalization into governed structures
- preservation of provenance and validation
- strict lifecycle containment at entry

Unlike Path 1 and Path 2:

- Path 3 does not produce training-ready datasets
- it does not demonstrate lifecycle promotion
- and it does not authorize dataset use

5.7 Role in the Overall System

Within the full system:

- Path 1 → internal reference training
- Path 2 → governed lifecycle progression
- Path 3 → external onboarding and containment

Together, these paths show how:

- internal sources
- runtime signals
- and external data

can be handled within a single controlled framework.

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6. Comparative Summary and Final Interpretation

6.1 Comparative Summary of the Three Paths

Path	Entry Type	Lifecycle Outcome	Primary Role
Path 1 — TR corpus reference	Internal TR source material	STATE_1 (training-preparation)	Reference dataset + controlled validation
Path 2 — Source-first lifecycle	Access-Mode frozen bundle	STATE_1	Governed lifecycle pathway
Path 2 — Runtime option	Runtime signal generation	STATE_2 (promotion held)	Runtime signal pathway
Path 3 — External bundle	External governed dataset	STATE_3	Controlled onboarding
Path 3 — Client dataset	Structured client input	STATE_3	Controlled onboarding

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6.2 System-Level Behavior

Across all validated runs, the system demonstrates:

- strict lifecycle gating
- controlled dataset progression
- separation between execution and authority
- preservation of provenance and traceability

A core principle applies:

execution produces artifacts, but does not confer authority

6.3 Path Roles Within the System

The three paths represent distinct system functions:

- Path 1 → reference dataset construction and validation
- Path 2 → lifecycle governance and promotion
- Path 3 → external onboarding and containment

Together, they demonstrate a unified system for:

- internal sources
- runtime signals
- external datasets

6.4 Path 2 — Access-Mode Release Clarification

The Path 2 lifecycle runs use updated Access-Mode artifacts:

- ID_2 operational artifacts
- ID_3 freeze artifacts

These updates produce:

- **four distinct session scenarios**

This is required because:

- lifecycle promotion requires **minimum candidate count = 4**

Earlier versions:

- produced only two sessions
- were insufficient for full promotion

The updated release enables:

- full lifecycle progression to STATE_1

6.5 Lifecycle Boundary Interpretation

Lifecycle progression is:

- explicit
- gated
- non-automatic

Key rules:

- STATE_1 does not imply dataset usage
- review does not imply approval
- structure does not imply correctness

6.6 Path 1 Training Clarification

For Path 1, a controlled training run was executed after STATE_1 preparation.

This run:

- validates dataset structure
- operates outside lifecycle state
- does not modify lifecycle outcome
- does not grant training authorization

This confirms:

- lifecycle governs readiness
- execution validates behavior
- authority remains external

6.7 Training and Authorization Boundary

Across the entire release:

No result should be interpreted as unrestricted fine-tuning authorization.

This applies to:

- Path 1 (reference dataset)
- Path 2 (lifecycle system)
- Path 2 runtime option
- Path 3 onboarding

All training activity is:

- controlled
- local
- non-authoritative

6.8 Final Release Interpretation

The v1.0 release represents:

- a **controlled, evidence-producing system**
- demonstrating governed dataset pathways

- without transferring authority into execution

It shows:

- datasets can be constructed and validated
- lifecycle progression can be enforced
- external data can be safely onboarded

while maintaining:

- containment
- traceability
- governance separation

6.9 Final Statement

The system is:

- operational, but controlled
- validated, but non-authoritative
- capable, but not self-approving

Execution produces structure.

Lifecycle governs progression.

Authority remains external.